IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

SENSOR SYSTEMS AND NETWORKS

Wednesday, 13 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

This is an OPEN book examination.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : W.T. Pike
Second Marker(s) : R. Syms

© Imperial College London

Special instructions for invigilators

This is an OPEN BOOK examination.

Special instructions for students

All questions should be answered.

The Answers

1. An autonomous vehicle moves in one dimension. Its state is

$$\mathbf{x}_k = \begin{bmatrix} p_k \\ v_k \end{bmatrix}$$

where p_k is position and v_k is velocity at time step k. The motion model is constant velocity driven by white acceleration noise $a_k \sim \mathcal{N}(0, q_a)$ sampled every Δt seconds:

$$\mathbf{x}_{k+1} = F\mathbf{x}_k + Ga_k$$

with

$$F = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, G = \begin{bmatrix} \Delta t^2/2 \\ \Delta t \end{bmatrix}.$$

A GPS sensor provides scalar position measurements

$$z_k = H\mathbf{x}_k + r_k, H = [1 \quad 0], r_k \sim \mathcal{N}(0, R).$$

At time $k-1$, the posterior estimate is

$$\hat{\mathbf{x}}_{k-1|k-1} = \begin{bmatrix} 100 \\ 5 \end{bmatrix}, P_{k-1|k-1} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

Take $\Delta t = 2$ s, $q_a = 0.25$, and $R = 9$.

a) Derive the discrete-time process-noise covariance matrix Q in terms of q_a and Δt.

[5]

b) Compute the predicted state mean $\hat{\mathbf{x}}_{k|k-1}$ and predicted covariance $P_{k|k-1}$.

[9]

c) At time k, a GPS measurement $z_k = 111$ is received. Compute:

i) the innovation.

[2]

ii) the innovation covariance,

[2]

iii) the Kalman gain,

[2]

iv) the updated state estimate $\hat{\mathbf{x}}_{k|k}$,

[2]

v) the updated covariance $P_{k|k}$.

[2]

d) A second position sensor is available, with measurement noise variance $R_d = 1$, but its data arrive with latency τ. Assume that delay can be approximated as an additional uncertainty growth $q_p\tau$ on position, with $q_p = 0.8\ \mathrm{m}^2/\mathrm{s}$. Derive a condition under which the delayed sensor is preferable to GPS for reducing current position uncertainty. Evaluate this condition for $\tau = 4$ s.

[6]

2. Communication in a sensor network is initiated using a *flood* from node *S*.

a) Explain what is meant by such a flood and how it produces a spanning tree with every reachable node in the network having a unique communication pathway to *S*.

[7]

b) A network of 7 nodes, *S*, *A, B, C, D, E, F* has the following edges with associated delays:

S-A: 2

S-B: 1

A-C: 2

A-D: 3

B-D: 1

B-E: 4

C-F: 2

D-F: 1

E-F: 3

i) Determine the time of first message reception for each sensor node assuming one time unit delay between reception and transmission.

ii) Draw the spanning tree for the network.

[6]

iii) Compare the total number of messages sent in the flood with that using the spanning tree to reach all the nodes from *S*.

[2]

c) Explain how the energy required to communicate can be further reduced after time synchronisation from the initial flood.

[6]

3. A company operates a sensor network and wants to learn a simple linear model in a federated manner. Each sensor node locally processes its data and shares only model parameters with a central server. The model has a single parameter w and predicts $\hat{y} = wx$ where x is a feature derived from sensor readings. There are two sensor nodes, each with a small local dataset:

Node 1: $(x, y) = (1, 2), (2, 4)$

Node 2: $(x, y) = (1, 1), (2, 2)$

The global model is initialized at $w^{(0)} = 0$. Each node performs one step of gradient descent on the loss $L(w)$, defined as mean square error of the prediction, with learning rate $\eta = 0.1$, starting from $w^{(0)}$, and sends its updated parameter to the central server. The server then performs federated averaging, weighted by the number of samples at each node.

a) Derive the gradient $\frac{dL}{dw}$ for this model.

[4]

b) Using this gradient and $\eta = 0.1$, compute the updated parameter after one step of gradient descent for each node.

[12]

c) Hence show that the global parameter $w^{(1)}$ after this one round of federated learning is 0.75.

[4]

d) For privacy, the network operator wants to protect individual readings using ε-differential privacy when releasing the aggregated parameter w. On the basis that the observations give an estimate of the numerical bounds of expected values, show that the scale parameter of the Laplace noise which should be added is $1/4\varepsilon$.

[4]

e)

i) Explain qualitatively how decreasing ε affects the amount of noise added and the accuracy of the released parameter in the context of this sensor network.

[3]

ii) In practice, why might we enforce clipping of node updates (e.g., capping the magnitude of each node's model update) before aggregation in a federated sensor network with differential privacy?

[3]